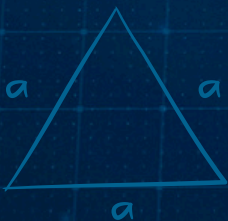




ALGEBRA NOTES

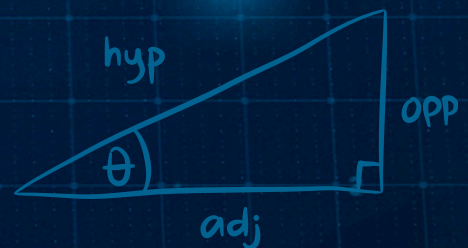


$$A = \frac{\sqrt{3}}{4} a^2$$



$$V = \frac{4}{3} \pi r^3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$\sin(\theta) = \frac{\text{opp}}{\text{hyp}}$$

A.P. Series, G.P. series & H.P

1) $T_n = An + B$ where $d = A$

2) $S_n = An^2 + Bn$ where $d = 2A$ (sum = quadratic equation)

3) $T_n = a + (n-1)d$

4) $S_n = \frac{n}{2} \{2a + (n-1)d\}$ or $\frac{n}{2} \{a + l\}$

To find T_n , we use three methods: —

for Case 1 & Case 2 (when $S_n = An^2 + Bn$): —

M1: —

put $n = 1, 2, 3, \dots$

$n = 1, T_1 = 8$

$n = 2, T_1 + T_2 = 14$

$d = 6$

M2: —

put $n = 1$ to find T_1

$An^2 + Bn \rightarrow d = 2A$

& solve

finding T_n when $S_n = An^2 + Bn + C$: —

use: — $T_n = S_n - S_{n-1}$ ✓ Ex: — $T_n = S_n - S_{n-1}$
 $T_{11} = S_{11} - S_{10}$

* If sum of 1st x numbers = sum of 1st y numbers
 Then, sum of 1st $(x+y)$ numbers = 0

* Double series: —

2, 5, 8, $1146 + 8n = T_n$ (Eq)

3, 5, 7, $1198 + 4n = T_y$ (Eq)

1) For finding 'd' → just look & find or $T_n = T_y$

2) For finding 'd' → LCM(d_1, d_2)

3) for finding no. of common terms: —

→ find $\min(l_1, l_2) = T_n$

→ find n from —

$T_n = a + (n-1)d$

4) for finding largest common term —

put n in the below eq ✓

$T_n = a + (n-1)d$

$$a, b, c, d \text{ are in A.P.} \quad \left| \quad a, b, c \text{ are in A.P.} \right.$$

$$\Rightarrow \boxed{a+d = b+c} \quad \left| \quad \boxed{b-a = c-b} \right.$$

$$A.M \Rightarrow b = \frac{a+c}{2} \quad \left[\text{where } a, b, c \text{ are in A.P.} \right]$$

$$1, \underbrace{x, y, z}_{AM}, 20 \Rightarrow \begin{aligned} &1) \text{ find } d \text{ from } T_n = a + (n-1)d \\ &2) \text{ find } (a+d), (a+2d), (a+3d) \end{aligned}$$

* Choosing: —

$$3 \text{ terms in A.P.} \rightarrow (a-d), a, (a+d) \quad [\text{Continuous}]$$

$$4 \text{ terms in A.P.} \rightarrow (a-3d), (a-d), (a+d), (a+3d) \quad [\text{Discontinuous}]$$

$$5 \text{ terms in A.P.} \rightarrow (a-2d), (a-d), a, (a+d), (a+2d)$$

$$* \begin{array}{l|l} 1, 3, 5, 7 & d_1 = 2 \\ 2, 4, 6, 8 & d_2 = 2 \\ 3, 7, 11, 15 & d_3 = 4 \end{array} \quad \left| \quad \begin{array}{l} 2, 5, 8, 11; d = 3 \quad (\text{Applicable for } \times, \div, +, -) \\ \downarrow \times 4 \\ 8, 20, 32, 44; d = 3 \times 4 = 12 \end{array} \right.$$

* G.P Series: —

$$\bullet \quad \boxed{T_n = a r^{n-1}}; \quad \boxed{r = \frac{T_2}{T_1}}; \quad S_n = \frac{a(1-r^n)}{1-r} \quad \checkmark$$

$$\bullet \quad S_n = \frac{a(r^n - 1)}{(r - 1)}$$

$$\bullet \quad S_n = \frac{a}{1-r} \quad [\text{when } -1 \leq r \leq 1] \rightarrow \text{sum} = \text{finite}$$

A.G.P (Arithmetic Geometric Progression)

$$1 \times 2^1 + 3 \times 2^2 + 5 \times 2^3 + 7 \times 2^4 \quad (\text{Example}) \Rightarrow d = 2, r = 2$$

Solved:- (A.G.P $\rightarrow$ G.P Convert) $\checkmark$

$$S = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + 5 \times \frac{1}{32} \quad ; \quad d = 1, r = 1/2 \quad [A.G.P]$$

$$S \times r = 1 \times \frac{1}{4} + 2 \times \frac{1}{8} + 3 \times \frac{1}{16} + \dots$$

$$S - S \times \frac{1}{2} = 1 \times \frac{1}{2} + 1 \times \frac{1}{4} + 1 \times \frac{1}{16} + 1 \times \frac{1}{32} + \dots \quad (\text{G.P series})$$

$$\Rightarrow S/2 = \frac{a}{1-r} = 1 \Rightarrow S = 2 \quad \checkmark$$

*) Sum of two AP (ratio given) $\rightarrow$ find ratio of n^{th} terms.

Then $\rightarrow \boxed{n = 2m - 1} \left\{ \begin{array}{l} \text{sum given} \\ \text{term not given} \end{array} \right\}$

*) Term given
To find sum $\Rightarrow \boxed{n = \frac{m+1}{2}} \left\{ \begin{array}{l} \text{term given} \\ \text{sum not given} \end{array} \right\}$

Difference A.P series: —

1, 2, 4, 7, 11, 16 $\therefore \boxed{T_n = an^2 + bn + c}$

$\underbrace{1}_1 \quad \underbrace{2}_2 \quad \underbrace{4}_3 \quad \underbrace{7}_4 \quad \underbrace{11}_5 \quad \underbrace{16}_6$

(frame equation) $\rightarrow n=1, T_1 = a + b + c = 1$
 $\rightarrow n=2, T_2 = 4a + 2b + c = 2$
 $\rightarrow n=3, T_3 = 9a + 3b + c = 4$

$\therefore a = \frac{1}{2}, b = -\frac{1}{2}, c = 1$

put these values $\rightarrow \boxed{T_n = \frac{n^2}{2} - \frac{n}{2} + 1}$ (we can find any term)

Sum of difference A.P: $\rightarrow \boxed{S_n = \sum_{n=1}^{20} T_n}$

Points to remember: —

$\sum_{n=1}^n n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

$\sum_{n=1}^n n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$\sum_{n=1}^n n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2$

$\left| \begin{array}{l} \sum_{n=1}^{20} \frac{n^2}{2} \\ = \frac{1}{2} \sum_{n=1}^{20} n^2 \text{ (rule)} \end{array} \right.$

Triangular series $\rightarrow 1, 3, 6, 10, 15, 21, \dots$
 (An example of difference AP series)

$\Rightarrow$ Triangular series $\rightarrow \boxed{T_n = \frac{n(n+1)}{2}}$

Difference G.P series: —

$$1, 3, 7, 15, 31$$

$$\underbrace{1 \quad 3}_{2} \quad \underbrace{3 \quad 7}_{4} \quad \underbrace{7 \quad 15}_{8} \quad \underbrace{15 \quad 31}_{16}$$

$$T_n = ar^{n-1} + bn + c$$

$$\Rightarrow r = 2$$

(frame equation)

$$\begin{aligned} n=1, T_1 &= a + b + c = 1 \Rightarrow a + b = 2 \\ n=2, T_2 &= 2a + 2b + c = 3 \Rightarrow 2a + b = 4 \\ n=3, T_3 &= 4a + 3b + c = 7 \end{aligned}$$

$$a = 2; b = 0; c = -1$$

$\therefore$ New equation $\rightarrow T_n = 2 \times 2^{n-1} + 0 \times n - 1$
after putting values $\Rightarrow T_n = 2^{n-1}$ ✓

Difference G.P sum: — $S_n = \sum_{n=1}^{20} T_n$ ✓

Sum of A.G.P with finite terms: —

$$S = 1 \times 3^1 + 3 \times 3^2 + 5 \times 3^3 + \dots + 39 \times 3^{20}$$

$$3S = 1 \times 3^2 + 3 \times 3^3 + 5 \times 3^4 + \dots + 39 \times 3^{21}$$

$$S - 3S = 1 \times 3^1 + 2 \times 3^2 + 2 \times 3^3 + \dots + 2 \times 3^{20} - 39 \times 3^{21}$$

(G.P series)

$$\Rightarrow -2S = 3 + 2 \times \frac{3(3^{20}-1)}{3-1} - 39 \times 3^{21} \text{ (Ans) } \checkmark$$

Remember: — (VVIP)

A.G.P $\rightarrow$ Step 1: — Subtract $S - S \times r$
 $\rightarrow$ Step 2: — Apply G.P formula $\left(\frac{a}{1-r} \text{ or } \frac{a(r^n-1)}{r-1} \right)$

Difference A.P $\rightarrow T_n = an^2 + bn + c$; find a, b & c

Difference G.P $\rightarrow T_n = ar^{n-1} + bn + c$; find a, b & c

Sum of Difference A.P/G.P $\rightarrow S_n = \sum_{n=1}^n T_n$

a, b $AM \Rightarrow \frac{a+b}{2}$ $GM \Rightarrow \sqrt{ab}$	a, b, c $AM \Rightarrow \frac{a+b+c}{3}$ $GM \Rightarrow \sqrt[3]{abc}$	$AM \geq GM$ (always) (for all +ve real numbers)
--	---	---

<u>Choosing in G.P.:</u> when product given $\rightarrow \frac{a}{x}, a, ax$ when sum given $\rightarrow a, ax, ax^2$	a, b, c are in G.P. Then $b = \sqrt{ac}$ $\Rightarrow b^2 = ac$ a, b, c are in A.P. Then, $b = \frac{a+c}{2}$
---	---

$T_n \rightarrow S_n$ (Conversion) [finding T_n & finding S_n]

$$2 \times 5 + 4 \times 7 + 6 \times 9 + \dots$$

$T_n = a + (n-1)d$ $= 2 + (n-1) \times 2$ $T_n \Rightarrow 2 + 2n - 2 = 2n$	$T_n = a + (n-1)d$ $= 5 + (n-1) \times 2$ $\Rightarrow 5 + 2n - 2$ $T_n \Rightarrow 2n + 3$
---	--

$$\therefore T_n = 2n \times (2n+3)$$

$$\& \boxed{S_n = \sum T_n}$$

↑ Remind Both ↓

Telescope Method (Sum of Series): — (Numerator)

$\sum_{d=1}^n (1 \times 2 \times 3 + 2 \times 3 \times 4 + 3 \times 4 \times 5)$ (when both d are same)	$S = \frac{1}{(n+1)d} [3 \times 4 \times 5 \times 6 - (3 \times 2 \times 1 \times 0)]$ where n is no. of terms in each term
--	--

(Denominator: —)

$$\frac{1}{1 \times 3 \times 5} + \frac{1}{3 \times 5 \times 7} + \dots + \frac{1}{21 \times 23 \times 25}$$

$$S = \frac{1}{(n-1)d} \left\{ \frac{1}{1 \times 3} - \frac{1}{23 \times 25} \right\}$$

Here $n = 3$ as $1 \times 3 \times 5$ (3 terms)
 $d = 2$

Numerator: —

$\hookrightarrow$ Extra term include ✓
 $\hookrightarrow$ Last - first
 $\rightarrow \frac{1}{(n+1)d}$
 Denominator — $\frac{1}{(n-1)d}$
 $\hookrightarrow$ first - last
 $\hookrightarrow$ Exclude term ✓

H.P (Harmonic Progression) $\Rightarrow$ Reverse of A.P

1, 2, 3, 4, 5 (A.P)

$\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ (H.P)

$$HP \rightarrow \frac{1}{AP} = \frac{1}{a+(n-1)d} = T_n$$

If a, b, c are in H.P

Then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

$$\Rightarrow b = \frac{2ac}{a+c}$$

$$T_n \text{ of H.P } (T_n) = \frac{1}{a+(n-1)d}$$

$$HM = \frac{2ab}{a+b} \text{ (If } a, b \text{ are in HP)}$$

Remember: — $AM \geq GM \geq H.M.$

Also Remember: — $AM \times HM = GM^2$

* Quadratic eqn Solving tips & tricks: —

=1: — $ax^2 + bx + c = 0$ $b = \square + \square$ (All positive) a, b, c

=2: — $ax^2 - bx + c = 0$ $b = \square + \square$ ($b = -ve, c = +ve$)

=3: — $ax^2 + bx - c = 0$ $b = \square - \square$ ($c = -ve$)

=4: — $ax^2 - bx - c = 0$ $b = \square - \square$ ($c = -ve$)

$c = -ve \Rightarrow \square - \square$ (Break in subtraction)

$c = +ve \Rightarrow \square + \square$ (Break in addition)

Max - Min (Important tricks & Numericals)

If Sum given: —

$$2a + 3b + c = 9$$

find max(abc)

$$\left(\frac{2a+3b+c}{3}\right) \geq (2a \times 3b \times c)^{1/3}$$

$$\Rightarrow abc \leq 9.5$$

sum gets divided by the powers

$$2a + 3b + 4c = 30; \text{ find } (a^2 b^3 c^2) \text{ max}$$

$$a, a, b, b, b, 2c, 2c$$

Apply AM $\geq$ GM

$$\left(\frac{a+a+b+b+b+2c+2c}{7}\right) \geq (a \times a \times b \times b \times b \times 2c \times 2c)^{1/7}$$

* Better method (when sum given): —

$$x + 2y + 3z = 6; \text{ find max}(x^1 y^2 z^3)$$

$$\frac{x}{1} = \frac{2y}{2} = \frac{3z}{3} = \frac{x+2y+3z}{1+2+3}$$

$$\left[\begin{array}{l} x | x^1 \Rightarrow \frac{x}{1} \\ 2y | y^2 \Rightarrow \frac{2y}{2} \\ 3z | z^3 \Rightarrow \frac{3z}{3} \end{array} \right]$$

$$\Rightarrow x = y = z = \frac{6}{6} = 1$$

$$\therefore x^1 y^2 z^3 = 1$$

If $36x + \frac{1}{x} \geq 12$ (for $x = +ve$ real)
then, $36x + \frac{1}{x} \leq -12$ (for $x = -ve$ real)

If Product given: — (sum gets divided by powers)

$$a^1 b^3 c^4 = 30; \text{ find min}(2a + 3b + 4c)$$

$$2a, b, b, b, c, c, c, c$$

$$AM \geq GM \Rightarrow \left(\frac{2a+b+b+b+c+c+c+c}{8}\right) = (2a b^3 c^4)^{1/8}$$

$$\Rightarrow \frac{2a+3b+4c}{8} = (2 \times 30)^{1/8} \text{ (Ans)}$$

Important variety: — $2a + 5b = 7$; find $(a+1)^2(b+2)^3$

$$\text{Let } a+1=x \mid b+2=y \mid 2a+5b=7 \Rightarrow 2x+5y=19 \quad \checkmark$$

$$\text{find } \rightarrow x^2 y^3$$

$$\frac{2x}{2} = \frac{5y}{3} = \frac{2x+5y}{5}$$

$$x = \frac{5y}{3} = \frac{19}{5}$$

$$\therefore x = \frac{19}{5}; y = \frac{19}{5} \times \frac{3}{5} = \frac{57}{25}$$

Max-Min (find Maximum & minimum values) [VVIP]

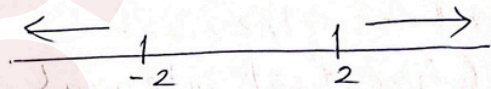
1) Product given $\rightarrow$ (Sum)_{min} $\rightarrow$ $AM \geq GM$ (method 1)

2) Sum given $\rightarrow$ (Product)_{max} $\rightarrow$ $\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$ (method 2)

* <u>Remind</u> $\rightarrow$ Sum given $\rightarrow$ ratio rule ($\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$)	S $\rightarrow$ R
Product given $\rightarrow$ AMGM rule ($AM \geq GM$)	P $\rightarrow$ AMGM

Remind $\rightarrow$ $x + \frac{1}{x} \geq 2$ (for +ve Real numbers) [VVIP]

$\hookrightarrow x + \frac{1}{x} \leq -2$ (for -ve Real numbers)



* Most Important rule $\rightarrow$ $AM \geq GM$ (for finding Min) ✓
Max

Every Numerical $\rightarrow$ try with $AM \geq GM$ for finding Max-Min

Udaan Concepts :-

1) $A^B = 1$

Case 1 :- when $B = 0$, check $A \neq 0$

Case 2 :- when $A = 1$, $B = \text{any value}$

Case 3 :- when $A = -1$, check $B = \text{even}$

2) $V + \frac{1}{W + \frac{1}{X + \frac{1}{Y}}} = \frac{222}{155}$ k(kt)

$$\Rightarrow V + \frac{1}{k} = 1 + \frac{67}{155}$$

Comparing $V = 1$, $\frac{1}{k} = \frac{67}{155} \Rightarrow k = \frac{155}{67}$ and so on...

Modulus & Inequality (Inequality)

Real No:— Can be both +ve & -ve as well as decimal

-ve Real no:— $-10, -5.5, -3$ etc

+ve Real no:— $+2, +2.5, +9$ etc.

Integer $\rightarrow$ Can be both +ve & -ve but not decimal

Example:— $-10, -9, -3, 2, 3, 8$, etc.

Whole no:— $0, 1, 2, 3, \dots$

Natural no:— $1, 2, 3, \dots$

GIF (greatest integral function)

$\rightarrow N = I + F$ (Representation)

Ex:— $[2.3] \rightarrow 2$ | $[-1.5] \rightarrow -2$

$$(1) \quad 2.3 = \underbrace{2}_{\checkmark} + 0.3$$

$$(2) \quad -1.5 = \underbrace{-2}_{\checkmark} + 0.5$$

$(1, 2) \rightarrow$ all values b/w 1 & 2.

$\{1, 2\} \rightarrow$ only 1 & 2.

$[1, 2] \rightarrow$ all values b/w 1 & 2 including 1 & 2.

Rule 1:— $x^2 = x$

Never
cancel
variables
on either
sides

$$\Rightarrow x^2 - x = 0$$

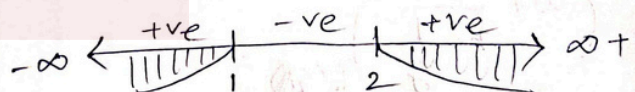
$$\Rightarrow x(x-1) = 0$$

$$\therefore x = 0, 1$$

Rule 2:—

$$(x-1)(x-2) \geq 0$$

Critical points $\rightarrow x = 1, 2$



Rule 3:— Coefficient of highest power must be positive always.

$$-x^2 + 3x \geq 0$$

$$\Rightarrow x^2 - 3x \leq 0$$

Coefficient of highest power = +ve always

$>$ or $< \Rightarrow ()$ Excludes

$\geq$ or $\leq \Rightarrow []$ Includes

Rule 4:— $-5x > 0$

$$\Rightarrow 5x < 0$$

multiply with $(-1) \Rightarrow$ Sign changes

Rule 5: — we can ignore even power when required: —

$$\begin{array}{l|l}
 n^{81} \times (n-1) \geq 0 & (n-1)^{31} (n-2)^{30} n^{81} \geq 0 \\
 \Rightarrow \underbrace{n^{80}} \times n(n-1) \geq 0 & \Rightarrow \underbrace{n^{80}} \times n \times \underbrace{(n-2)^{30}} \times (n-1) \times \underbrace{(n-1)^{30}} \geq 0 \\
 \Rightarrow n(n-1) \geq 0 & \Rightarrow n(n-1) \geq 0
 \end{array}$$

But we must check at $n=0$

But check also at $n=0, n=1, n=2$

Rule 6: —

$$\frac{n}{(n-1)} \geq 0 \Rightarrow \frac{n(n-1)}{(n-1)^2} \geq 0 \Rightarrow n(n-1) \geq 0 \times (n-1)^2$$

$$\Rightarrow n(n-1) \geq 0 \quad \left| \text{we can only shift positive terms to another side} \right.$$

Rule 7: —

$$\frac{n}{n-2} \geq 2$$

Always transfer constants
to the same side

$$\Rightarrow \frac{n}{n-2} - 2 \geq 0$$

Inequality $\begin{cases} \nearrow \text{Critical points} \rightarrow \text{Number Line Plot} \\ \searrow \text{Range} \leftarrow \end{cases}$

Modulus

Rule 1: — $|x| = x$ if $x \geq 0$ } mod opening
(value of variable) $= -x$ if $x < 0$ } wrt. critical point

Ex: — $|x+2| = x+2$ if $x \geq -2$ } mod opening
critical point $= -(x+2)$ if $x < -2$ } wrt critical point
 $\hookrightarrow x+2=0$
 $\Rightarrow x=-2$ $\xrightarrow{- (x+2) \quad -2 \quad (x+2)}$ (Always open mod at critical points)

Rule 2: — $|x| = a$
 $\Rightarrow x = \pm a$

Ex: — $|x-5| = 3$
 $\Rightarrow x-5 = \pm 3$
 $\Rightarrow x = 5 \pm 3$ $\begin{cases} 8 \text{ (Ans)} \\ 2 \text{ (Ans)} \end{cases}$

Rule 3: — $|x| \leq a$
 $\Rightarrow -a \leq x \leq a$

Rule 4: — $|x| \geq a$
Then, $x \geq a$ & $x \leq -a$

Multiple Critical point: —

$$|x-1| + |x-5| = 3 ; \text{ critical points } = 1, 5$$



Make 3 cases:

Case 1: — $x \geq 5$, Case 2: — $-1 < x < 5$, Case 3: — $x \leq -1$
Open mod of the original equation & solve.

$\hookrightarrow$ Assume any value & check
if +ve $\rightarrow$ +ve open
if -ve $\rightarrow$ -ve open

Indices Rules: — 1) $a^x \times a^y = a^{x+y}$

2) $a^x \times b^x = (ab)^x$ 3) $(a^b)^x = a^{bx}$

4) $\frac{a^x}{a^y} = a^{x-y}$ 5) $a^{b^x} = a^{b^x}$

6) $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$ 7) $\sqrt[a]{b} = b^{1/a}$

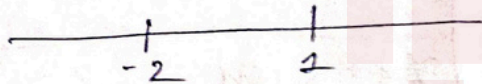
$$(2^5)^3 = 2^{15} \text{ (Bracket } \rightarrow x)$$

$$2^{5^3} = 2^{125} \text{ (No Bracket } \rightarrow \text{power)}$$

(Remind these)

$$Q) (|x-1|-4)(|x+2|-5) < 0$$

Critical point: - $x-1=0$, $x+2=0$
 $\Rightarrow x=1$ $\Rightarrow x=-2$



Make 3 cases & solve

- Rule 4: - $||x-3|| = 5$

Always assume the innermost mod as a [VVIP]

Let $|x-3| = a$ (Applicable for double/triple mods etc.)

Now $|a| = 5$

$\Rightarrow a = \pm 5$

$|x-3| = 5$

$\Rightarrow x-3 = \pm 5$

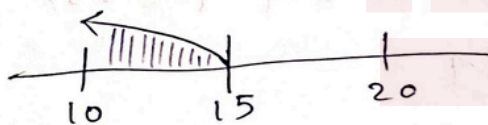
$\Rightarrow x = 3 \pm 5$

$|x-3| = -5$

Not possible (value of mod can't be negative)

Rule 5: - Comparison of mod : $\Rightarrow$

$|x-10| \leq |x-20|$



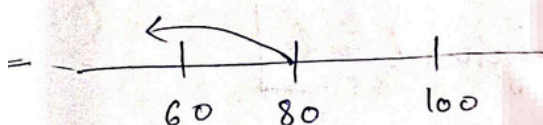
1) Calculate middle of Critical value

2) $|x-10| \leq |x-20|$

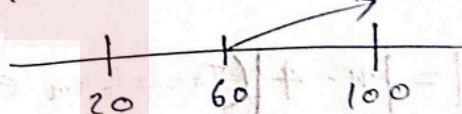
direct arrow towards $|x-10|$ from 15
 jake chote kote hobe, tar dake arrow
 jisko chote karana hoga, uski tarah arrow

- Ex: - $|x-60| < |x-100| < |x-20|$

$|x-60| < |x-100|$



$|x-100| < |x-20|$



Thus solution (combine both cases)

$60 < x < 80$ (Ans)

Rule 6:- finding minimum values of mod additions

* $|x-1| + |x-3| + |x-7|$

Critical points $\rightarrow x = 1, \underline{3}, 7$ ($n = \text{odd}$)

Median = 3

Putting 3 in the equation

$$|3-1| + |3-3| + |3-7|$$

$$= 2 + 0 + 4 = 6 \text{ (min value)}$$

if $n = \text{odd}$

median / middle

= $\frac{n+1}{2}$ th term ✓

if $n = \text{even}$

median / middle

= $\frac{n}{2}$ th term and

$\frac{n}{2} + 1$ th term

* $|x-3| + |x-5| + |x-7| + |x-20| + |x-25| + |x-28|$

Critical points $\rightarrow x = 3, 5, 7, \underline{20}, \underline{25}, 28$ ($n = \text{even}$)

Median $\rightarrow 7$ and 20

Putting any term from 7 to 20 will yield min value ($7 \leq x \leq 20$)

$$|7-3| + |7-5| + |7-7| + |7-20| + |7-25| + |7-28|$$

$$= 4 + 2 + 0 + 13 + 18 + 21$$

$$= 58 \checkmark$$

$$|14-3| + |14-5| + |14-7| + |14-20| + |14-25| + |14-28|$$

$$= 11 + 9 + 7 + 6 + 11 + 14$$

$$= 58 \checkmark$$

Transform eqn:- $x^2 - 5|x| + 4 = 0$

$$\Rightarrow |x|^2 - 5|x| + 4 = 0 \text{ (Transform)}$$

$$\Rightarrow a^2 - 5a + 4 = 0 \text{ (assume)}$$

* $|4-x| = |x-4|$ (Transform eqn) [General Rule]

Logarithm:

1) $\log_a b \Rightarrow a > 0, b > 0, a \neq 1$ (Condition)

2) $\log_a b = \frac{\log b}{\log a} = \frac{\log_{10} b}{\log_{10} a}$

*) $\log_b a = \frac{\log_c a}{\log_c b}$

3) $\log_a b = \frac{1}{\log_b a}$

*) $-\log_b a = \log_b \frac{1}{a}$ or $\log_{\frac{1}{b}} a$

4) $\log_a (mn) = \log_a m + \log_a n$

5) $\log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$

6) $\log_a m^n = n \log_a m$; $\log_a n^m = \frac{1}{n} \log_a m$

7) $\log_a a = 1$; $\log 1 = 0$

8) $\log_a b = x$ (VVIP) [inverse property]
 $\Rightarrow b = a^x$

9) $a^{\log_c b} = b^{\log_c a}$ (Interchange property)

10) $a^{\log_a m} = m$ (VVIP)

11) $\log_a m^x = \frac{x}{m}$

If $\log A, \log B, \log C$ are in A.P.

Then, $B^2 = AC$

Then, A, B, C are in G.P.

(Proved by Logic)

12) $\frac{\log_a b}{\log_a c} = \log_c b$

*) How many digits in 2^5 (Formula = Char + 1)?

14) $\log 2 = 0.301$

$\log 3 = 0.477$

$\log 5 = 0.699$

$[\log 2^5] + 1$

$= [5 \log 2] + 1$

$= [1.505] + 1$

$= 1 + 1 = 2 \therefore$ Two digits in 2^5

*) $\log x = 2.9$

Char = 2 (integral $\rightarrow +, -, 0$)

Mantissa = +0.9 (Fraction / 0 (always +ve))

$\log x = -1.5$

Char = -2

Mantissa = +0.5

Function)

1) $F(x) = x + 5$

We can put any value of x : $F(1) = 1 + 5 = 6$

2) $F(x+2y, x-2y) = 7x+5y$

find $F(3, 2)$ —

Then, $x+2y = 3$

$x-2y = 2$

& solve by putting value of x & y to get the solution

$F(x+2y, x-2y) = 7x+5y$

find $F(x, y)$

Then assume —

$x+2y = a$

$x-2y = b$

find value of x & y & solve

* Put values in function $F(x)$ & check for cyclicity ✓

3) $f(x+1) = f(x) - f(x-1)$ | cyclicity of 6

Ex:- $f(5) = f(4) - f(3)$ | 1, 7, 6, -1, -7, -6

Same as $f(t) = f(t-1) - f(t-2)$ ↑

* In this above case, sum of any consecutive terms is 0.

$1 + 7 + 6 + (-1) + (-7) + (-6) = 0$

If $f(x) = 9$
 $f(f(x)) = 9$
 $f(f(f(x))) = 9$
 (Concept)

Concept:-

$f(0) = f(1) = f(2) = \text{same value}$

It is called constant value function.
 or Constant function.

* $3f(x) + 2f(-x) = 5x - 10$

find $-f(1)$?

Ans:- Substitute x with $-x$ ($x \rightarrow -x$)

$3f(x) + 2f(-x) = 5x - 10 \dots \text{eq(1)}$

$3f(-x) + 2f(x) = -5x - 10 \dots \text{eq(2)}$

Then put $x = 1$ & solve

* $2f(x) + 3f(\frac{1}{x}) = 5x + 2$

find $-f(2)$

Ans:- Replace x by $\frac{1}{x}$ ($x \rightarrow \frac{1}{x}$)

$2f(x) + 3f(\frac{1}{x}) = 5x + 2 \dots (1)$

$2f(\frac{1}{x}) + 3f(x) = \frac{5}{x} + 2 \dots (2)$

Then, put $x = 2$ & solve

Quadratic equation

polynomial $\rightarrow a_1x^n + a_2x^{n-1} + \dots + a_nx^{n-n}$

max power (n) is called degree; degree = no. of roots

Quadratic $\rightarrow ax^2 + bx + c = 0$; roots $\rightarrow \alpha, \beta$

degree = 2; no. of roots = 2

roots satisfy the equation completely putting $x = \alpha$ or β

$$ax^2 + bx + c = 0$$

$$\rightarrow x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} (\alpha)$$

$$\rightarrow x = \frac{-b - \sqrt{b^2 - 4ac}}{2a} (\beta)$$

$$\alpha + \beta = -\frac{b}{a} \quad (\text{VVIP})$$

$$\alpha\beta = \frac{c}{a}$$

Cubic equation: — $ax^3 + bx^2 + cx + d = 0$

roots $\rightarrow \alpha, \beta, \gamma$; $\alpha + \beta + \gamma = -b/a$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -d/a$$

rule: —

-ve

+ve

-ve

+ve

Important Algebra formulae: —

$$1) (a-b)^2 = (a+b)^2 - 4ab$$

$$2) (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$3) (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$4) a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$5) a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$6) (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

biquadratic polynomial —

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

roots $\rightarrow \alpha, \beta, \gamma, \delta$

$$1) \alpha + \beta + \gamma + \delta = -b/a$$

$$2) \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c/a$$

$$3) \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a$$

$$4) \alpha\beta\gamma\delta = e/a$$

Nature of roots

$$\text{Discriminant (D)} = b^2 - 4ac$$

1) when $D > 0$ ————— $\left[\begin{array}{l} \text{Real \& distinct roots} \\ \text{Unequal roots} \end{array} \right]$

$$x = \frac{-b + \sqrt{D}}{2a}, x = \frac{-b - \sqrt{D}}{2a}$$

2) when $D = 0$ —————

$$x = -\frac{b}{2a} \left[\begin{array}{l} \text{Real \& equal roots} \\ \text{Unique solution} \end{array} \right]$$

Real & unequal — $D > 0$

Real & equal — $D = 0$

Real $\rightarrow D \geq 0$

Imaginary $\rightarrow D < 0$

3) when $D < 0$ —————

$$x = \frac{-b + \sqrt{D}}{2a}$$

$$x = \frac{-b - \sqrt{D}}{2a} \left[\begin{array}{l} \text{Imaginary roots} \\ \text{\& Unequal roots} \\ \text{(distinct)} \end{array} \right]$$

unequal roots: —

$D > 0$ or $D < 0$

* $ax^2 + bx + c = 0$

If a, b, c are rational values & D is a perfect square
 $\rightarrow$ roots will be rational

* $ax^2 + bx + c = 0$

If a, b, c are rational values & D is not a perfect square
 $\rightarrow$ roots are irrational & impairs $(2 + \sqrt{3}), (2 - \sqrt{3})$

* Conversion into a perfect square: —

1) Make the coefficient of x^2 as 1

2) Make the coefficient of x as half $\rightarrow$ square it (add & subtract)

Ex: — $x^2 + 8x + 2$

$$\Rightarrow x^2 + 8x + \left(\frac{8}{2}\right)^2 - \left(\frac{8}{2}\right)^2 + 2$$

$$\Rightarrow x^2 + 8x + 16 - 16 + 2$$

$$\Rightarrow (x+4)^2 - 14 \text{ (Ans)}$$

Ex: — $3x^2 + 12x + 1$

$$\Rightarrow 3(x^2 + 4x + \frac{1}{3})$$

$$\Rightarrow 3(x^2 + 4x + 2^2 - 2^2 - \frac{1}{3})$$

$$\Rightarrow 3((x+2)^2 - 4 + \frac{1}{3}) \checkmark$$

$$*) y = ax^2 + bx + c$$

$$\Rightarrow y + \frac{D}{4a} = a \left(x + \frac{b}{2a} \right) \quad [\text{by Derivation}]$$

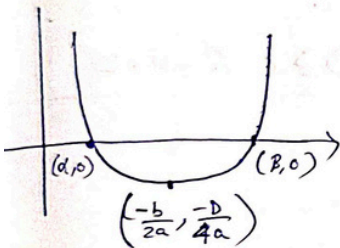
$\therefore$ Its a parabolic graph $y = ax^2$

$$\text{At } \boxed{x_{\min} = -\frac{b}{2a}}, \boxed{y_{\min} = -\frac{D}{4a}} \quad (\text{VVIP})$$

$$y = ax^2 + bx + c$$

Case 1: $- \boxed{a > 0}$ (opening upwards):

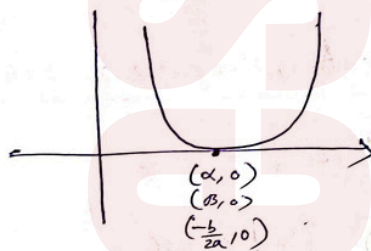
S1: $- \boxed{D > 0}$



$$\boxed{y = +ve, -ve \text{ or } zero}$$

$$\begin{aligned} x < \alpha &\rightarrow y > 0 \\ x > \beta &\rightarrow y > 0 \\ x = \alpha \text{ or } \beta &\rightarrow y = 0 \\ \alpha < x < \beta &\rightarrow y < 0 \end{aligned}$$

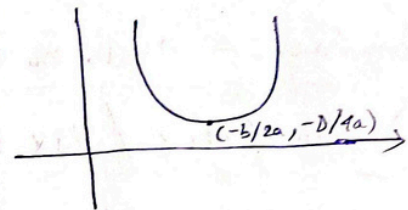
S2: $- \boxed{D = 0}$



$$\boxed{y \geq 0}$$

$$\begin{aligned} x < \alpha &\rightarrow y > 0 \\ x > \alpha &\rightarrow y > 0 \\ x = \alpha &\rightarrow y = 0 \end{aligned}$$

S3: $- \boxed{D < 0}$

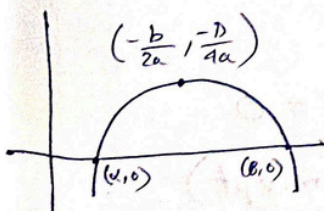


$$\boxed{y > 0}$$

Always positive $\checkmark$
(VVIP)

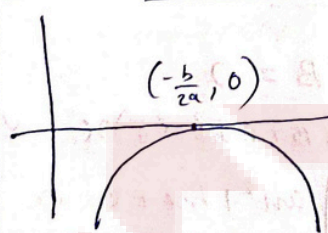
Case 2: $- \boxed{a < 0}$ (opening downwards):

S1: $- \boxed{D > 0}$



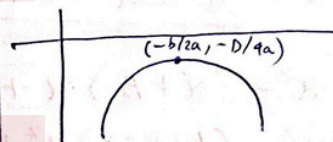
$$\boxed{y = +ve, -ve, 0}$$

S2: $- \boxed{D = 0}$



$$\boxed{y \leq 0}$$

S3: $- \boxed{D < 0}$



$$\boxed{y < 0}$$

$$*) f(x) > 0 \Rightarrow \boxed{D < 0}$$

$$f(x) \geq 0 \Rightarrow \boxed{D \leq 0}$$

* $f(n) \geq 0$ & $f(m) = 0 \Rightarrow \boxed{D = 0}$
 $\Rightarrow f(n) = a(n-m)^2$

* Max & Min value of Rational : —

F1: — $y = \frac{n^2 + n + 1}{n^2 - n + 5}$; Cross multiplication $\Rightarrow an^2 + bn + c = 0$
 $\boxed{D \geq 0}$ as $n \in \mathbb{R}$

F2: — $y = \frac{n}{n^2 + 8n + 7}$; divide by n || Cross multiplication
 $AM \geq GM$ $D \geq 0$ as $n \in \mathbb{R}$

F3: — $y = \frac{n^2 - 8n + 20}{4 - n}$; Perfect square || Cross multiplication
 $AM \geq GM$ $D \geq 0$

F4: — $y = \frac{n^2 + n + 7}{n^2 + 4 - 6}$; Cross multiplication $D \geq 0$ ✓

Rambaran: — Cross multiplication, Compare $an^2 + bn + c$, $D \geq 0$ & solve

* Min of Max / Max of Min : —

Q) $f(n) = \min(n-2, 4-n)$ find max.

$n-2 = 4-n$ (equate)

$\Rightarrow 2n = 6$

$\Rightarrow n = 3$

put value $n = 3$

$n-2 = 3-2 = 1$

$4-n = 4-3 = 1$

$\therefore \max = 1$

Step 1: — equate the terms

Step 2: — put the value of n in those terms

(Same steps to be followed while Calculating Min of Max)

* Root to equation Conversion : —

$\alpha, \beta \rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$\alpha, \beta, \gamma \rightarrow x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \alpha\gamma)x - \alpha\beta\gamma = 0$

Ex: — 2, 3 are roots

$\Rightarrow x^2 - 5x + 6 = 0$

$ax^2 + bx + c = 0$ (Concept)

$\Rightarrow \boxed{a(x-\alpha)(x-\beta) = 0}$ ✓

Concept: — If roots are of opposite sign (Ex: — +5, -2)

Then $\boxed{\alpha\beta < 0}$ ✓

Advance concepts (Additional)

* Two Linear Equation:-

$$a_1x + b_1y + c_1 = 0 \rightarrow m_1 = -\frac{a_1}{b_1}$$

$$a_2x + b_2y + c_2 = 0 \rightarrow m_2 = -\frac{a_2}{b_2}$$

1) unique solution $\rightarrow \boxed{\frac{a_1}{a_2} \neq \frac{b_1}{b_2}} \text{ (VVIP)}$

2) not unique $\rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2}$

[necessary conditions]

Parallel case $\rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
(0 solutions)

Superimpose case $\rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
(infinite solutions)

* How to find Common roots?

$$a_1x^2 + b_1x + c_1 = 0$$

$$- a_2x^2 + b_2x + c_2 = 0 \quad \left[\text{Case 1:- exactly 1 common root} \right]$$

find value of x (subtraction method)

put the value of x in both equation $\rightarrow$ if satisfy $\rightarrow$ common root ($\checkmark$)
else $\rightarrow$ false root ($\times$)

OR $(a_2b_2 - a_1b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$

if the above is true, we can find any value (eqn method)

Case 2:- If both roots are common.

$$a_1x^2 + b_1x + c_1 = 0$$

$$a_2x^2 + b_2x + c_2 = 0 \quad \text{then, } \boxed{\frac{a_1}{b_1} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

for cubic equation (3 common roots):

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{d_1}{d_2}$$

*) finding the minimum value of equations : —

Ex:- $x^2 + 4xy + 6y^2 - 4y + 4 = 0$

Ex:- $x^2 + 9y^2 + x - 6y = -5/4$

Ex:- $u^2 + (u - 2v - 1)^2 = -4v(u + v)$

Method 1 :- Convert them into perfect square

Method 2 :- Partial differentiation w.r.t to x & w.r.t. to y.

find — $\frac{f(x, y)}{f x}$ & $\frac{f(x, y)}{f y}$

*) $x + \frac{1}{x} = 1$	$x + \frac{1}{x} = -1$	$x + \frac{1}{x} = \sqrt{3}$
$\Rightarrow x^3 = -1$	$\Rightarrow x^3 = 1$	$\Rightarrow x^6 = -1$
$\Rightarrow x^6 = 1$	$\Rightarrow x^6 = 1$	(Remind these)

*) $x + \frac{1}{x} = 4$

Squaring we get,	Cubing we get —
$x^2 + \frac{1}{x^2} = 14$	$x^3 + \frac{1}{x^3} = 52$

How to find $x^5 + \frac{1}{x^5}$?

Ans:- $(x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) = 14 \times 52$

$\Rightarrow x^5 + \frac{1}{x^5} + x + \frac{1}{x} = 728$

$\Rightarrow x^5 + \frac{1}{x^5} = 724$ (Ans)

* Area of Modulus

1) $|x| + |y| = k$; Area = $2k^2$

2) $|x \pm a| + |y \pm b| = k$; Area = $2k^2$

3) $|x+y| + |x-y| = k$; Area = k^2

4) $|px \pm a| + |qx \pm b| = k$; Area = $\frac{2k^2}{pq}$

5) $|px| + |qx - y| = k$; Area = $2k^2/pq$

6) $|x| + |x-y| = k$; Area = $2k^2$

$|x+y| + |x-y| = k$; Area = k^2

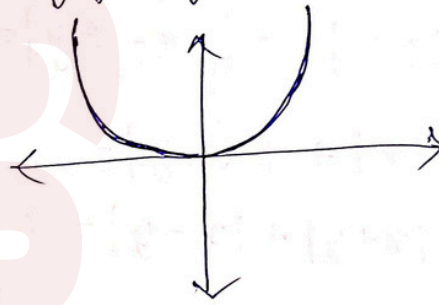
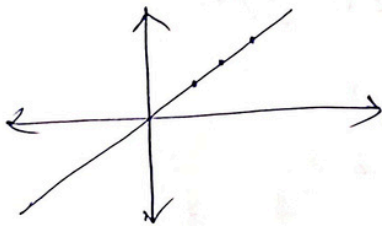
1) Graphs & Area

$f(x)$ is the representation of a function

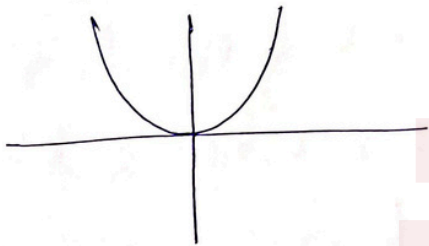
if $f(-x) = -f(x) \Rightarrow$ odd function (symmetrical $\rightarrow$ origin)

if $f(-x) = f(x) \Rightarrow$ even function (symmetrical $\rightarrow$ y-axis)

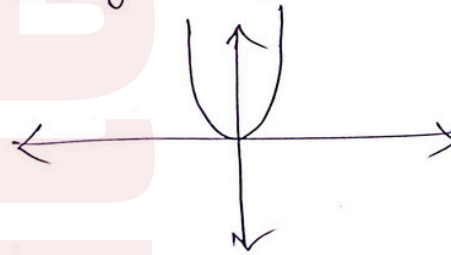
1) Graph of $y = x$ (odd function) | 2) Graph of $y = x^2$ (Even function)



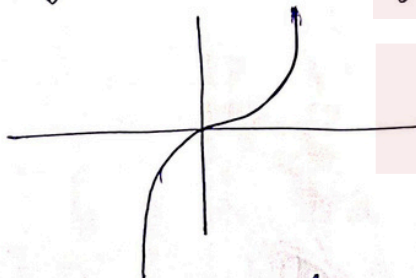
(3) $y = x^4$ (even fn)



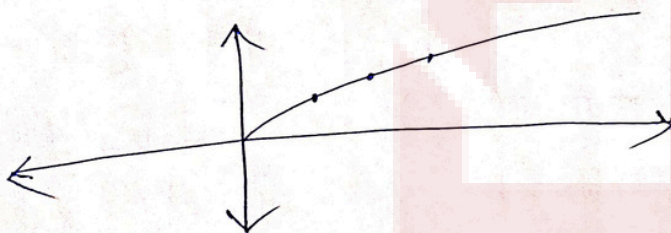
(4) $y = x^6$ (even fn)



(5) $y = x^3$ (odd function) : —

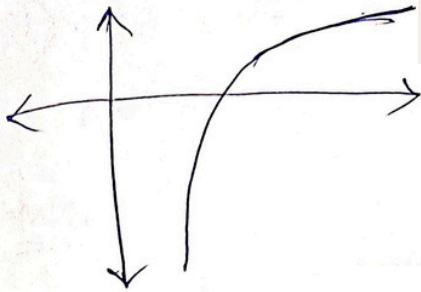


(6) $y = \sqrt{x}$ (neither even, nor odd) : —

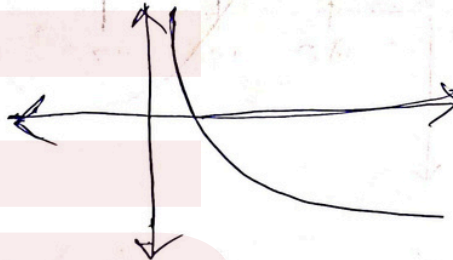


(7) $y = \log_a x$: — $\begin{cases} 0 < a < 1 : - \text{decreasing fn} \\ a > 1 : - \text{increasing fn} \end{cases}$

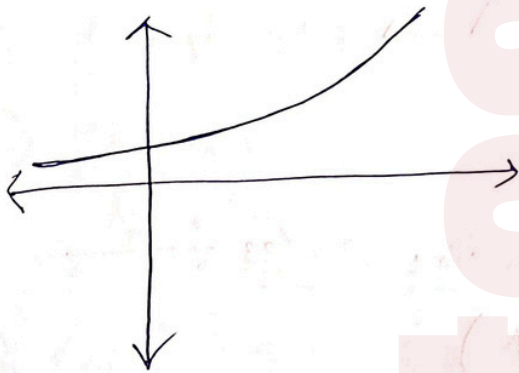
when $a > 1$: —



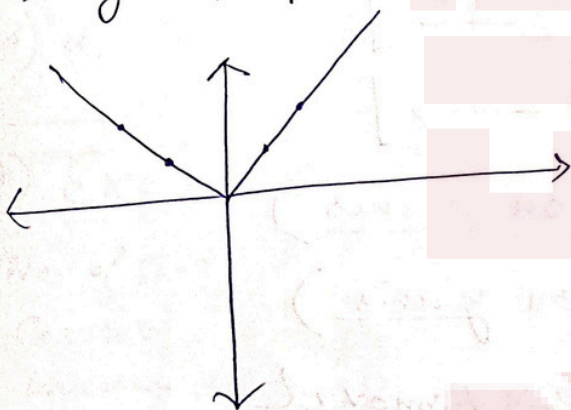
when $0 < a < 1$: —



(8) $y = e^x$ (increasing fn) $\begin{cases} x \rightarrow \infty \\ x \rightarrow -\infty \\ x \rightarrow 0 \end{cases}$



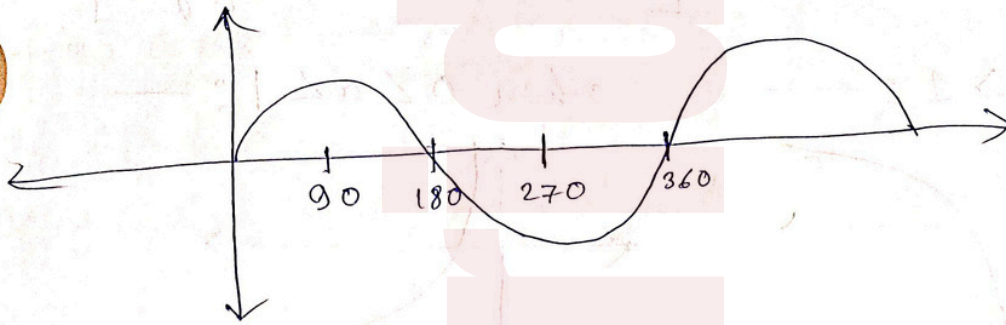
(9) $y = |x|$



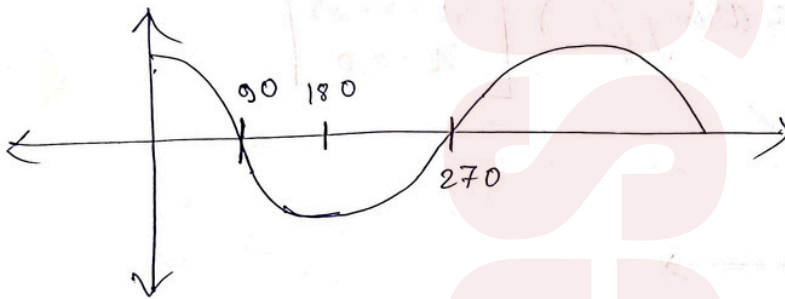
(10) $|y| = e^x$
 $\Rightarrow y = \pm e^x$



12) $\sin x$ (odd function) : —



13) $\cos x$ (even function) : —



* $y = f(x)$ graph construction rules :

- 1) $y = f(x) + a$ (up shift $\uparrow$)
- 2) $y = f(x) - a$ (down shift $\downarrow$)
- 3) $y = f(x + a)$ [Left shift $\leftarrow$]
- 4) $y = f(x - a)$ [Right shift $\rightarrow$]
- 5) $y = -f(x)$ (symmetry on x axis)
- 6) $y = f(-x)$ (symmetry on y axis)
- 7) $y = f(|x|)$ → 1) Negative x Removal
2) Construct symmetry w.r.t y axis
- 8) $y = |f(x)|$ → 1) Negative y Removal
2) Construct symmetry w.r.t x axis
- 9) $|y| = f(x)$ → symmetry on x axis.

Advance Concepts :

* Transformation of equation

1) $x^2 - 5x + 6 = 0$. The roots of the eqn are α, β
find the eqn whose roots are $\alpha+2, \beta+2$

M1 $x^2 + 3x + 1 = 0$

$\alpha + \beta = -3$

$\alpha\beta = 1$

$x^2 - (\alpha+2 + \beta+2)x + (\alpha+2)(\beta+2) = 0$

M2 (Trick $\checkmark$)

Just replace x
with $x-2$

(Reverse Assumption)

* Trick : (Reverse Assumption)

1) $\alpha, \beta \xrightarrow[\text{whose roots are}]{\text{find the eqn}} \alpha+2, \beta+2$ (Assume $x \xrightarrow{\text{as}} x-2$)

2) $\alpha, \beta \longrightarrow \alpha^2, \beta^2$ (Assume $x \rightarrow \sqrt{x}$)

3) $\alpha, \beta \longrightarrow \alpha-1, \beta-1$ (Assume $x \rightarrow x+1$)

4) $\alpha, \beta \longrightarrow \frac{1}{\alpha}, \frac{1}{\beta}$ (Assume $x \rightarrow \frac{1}{x}$)

* Exam question :

No. of question = 15

Correct answer (m) = +3

Wrong answer = -1

How many values are possible?

Ans: — max score = $15 \times 3 = 45$ | min score = $15(-1) = -15$

$\therefore$ No. of values (ideal) = 45 to $-15 = 61$ values

Scores not possible = $\frac{n(n-1)}{2} = \frac{3(3-1)}{2} = 3$ values
($n=1$ correct ans marks)

$\therefore$ No. of possible values = $61 - 3 = 58$ (Ans)

* Exam Concept (Long method): —

a = Total correct Ans

b = Total incorrect Ans

c = Total unattempted Questions

$$a + b + c = 15 \text{ (Total Question)}$$

A/Q — $3 \times a - b = k \text{ (score)}$

Correct $\rightarrow a = 15, b = 0, k = 45$

$a = 14, b = 0 \rightarrow 42$

$b = 1 \rightarrow 41$

$a = 13, b = 0 \rightarrow 39$

$b = 1 \rightarrow 38$

$b = 2 \rightarrow 37$

$a = 12, b = 0 \rightarrow 36$

$b = 1 \rightarrow 35$

$b = 2 \rightarrow 34$

$b = 3 \rightarrow 33$

⋮

Missing values

44, 43, 40

$\therefore$ No of missing values
= 3 ✓

Max score = 45

Min score = -15

ideal no of values = 61 ✓ (45 to -15)

— But, No. of missing values = 3

Total possible values = 58 possible values

#) Concept - 14) Product of irrational roots

$$(2n-1)(2n-3)(2n-5)(2n-7) = 9$$

$$\Rightarrow (2n-1)(2n-7) \times (2n-3)(2n-5) = 9$$

$$\Rightarrow \underbrace{(4n^2 - 16n + 7)}_a \times \underbrace{(4n^2 - 16n + 15)}_a = 9$$

$$\Rightarrow (a+7)(a+15) = 9 \quad \left| \text{Let } 4n^2 - 16n = a \right.$$

$$\Rightarrow a^2 + 22a + 96 = 0$$

$$\Rightarrow a = -6 \text{ or } -16$$

$$\text{when } a = -6$$

$$4n^2 - 16n = -6$$

$$\Rightarrow n = \frac{2 \pm \sqrt{10}}{2}$$

✓

$$\text{when } a = -16$$

$$4n^2 - 16n = -16$$

$$\Rightarrow n = 2$$

✗ (rational = eliminated)

$$\therefore \text{product} = \left(2 + \frac{\sqrt{10}}{2}\right) \left(2 - \frac{\sqrt{10}}{2}\right) \quad (\text{Ans})$$